

Vehicles- Petrol/Diesel/Electric

Various locations

PERSONAL PROTECTIVE EQUIPMENT



Version: 1	Key Contact: Eli Crawshaw
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HAZARDS ASSOCIATED WITH THIS EQUIPMENT

PERMANENT HAZARDS	List	Control Method	Lock out Yes or No
Electricity	Electric shock, electrical burn	Remind students and tutors to be situationally aware when working on the car electrical circuit, where necessary ensure the car is turned off and the battery is isolated. Teach fundamentals of how electricity works on running engines	
Moving parts	Physical injuries including tears, crushing and loss of limbs	Remind students and tutors to be situationally aware when working around vehicles and in the engine bay. Explain the importance of high-pressure lines and how to correctly work on them. Demonstrate and use correct starting procedures.	
Manual handling	Material Safety Data	Place vehicle at correct heights to remove components and use transmission jack to assist in removing and installing of components as required. Ensure hoist is in ergonomically correct position to work on.	
Sharp objects	Physical Injuries, cuts and splinters	Remind students and tutors to be situationally aware.	
Slips and trips	Physical injuries	Ensure good house keeping. Sweep, clean and tidy as you go, don't leave rubbish on ground, mop up spills etc.	
Noise	Hearing damage	Use in a well-ventilated area and wear appropriate PPE.	
Hazardous substances	Skin, eye, and respiratory irritation	Use in a well-ventilated area and wear appropriate PPE.	
Dust	Skin, eye, and respiratory irritation	Use in a well-ventilated area when using compressed air, blow outside where possible. Use appropriate PPE including masks.	
Medical conditions	Any person with medical condition including injury, epilepsy etc	Remind students and tutors NOT to operate a motor vehicle while on prescription medicine that could legally impair their alertness or judgment. Students are to inform tutors if they are on prescription medication.	
Hazardous materials	Skin, eye, and respiratory irritation, chemical burns	Use in a well-ventilated area and wear appropriate PPE.	
Fatigue	Crash, physical injury	Remind students and tutors NOT to operate a motor vehicle if run down or tired as it affects alertness or judgment.	
Flammable	Explosions or fire	Use in a well-ventilated area and wear appropriate PPE. Ensure correct draining procedure for all flammable liquids. Have fire extinguishers present. Indicate where they are located.	

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PURPOSE/PROCESS



PROCEDURES:

DO NOT use this machine unless a tutor has instructed and trained you in its safe use and operation

Pre-Start Check

1. DO NOT use this machine unless you have proven to your trainer that you are competent in its safe use and operation.
2. Locate and ensure you are familiar with all vehicle operations and controls.
3. Tutors are to ensure previous work is double checked and is in working order before students start working on the training vehicles.
4. Ensure the equipment you will be using to work on the car has been properly maintained.
5. If you are going to use a trolley jack and axle stands, ensure the ground is firm and on a level surface.
6. Ensure the area is clean and clear of grease and oil.
7. Check workspaces and walkways to ensure no slip/trip hazards are present.
8. Check all safety devices are in good condition.
9. Ensure the work area you will be working in is well-ventilated. Fit extraction system where appropriate and start the fume extraction unit before running the vehicle.
10. Ensure all flammable materials are correctly stored before operating.
11. Have a valid driver's license in the appropriate class for the motor vehicle being driven.
12. Do NOT operate a motor vehicle after consuming alcohol or other substances (including prescription and non-prescription medications) that could legally impair their alertness or judgment.
13. Do NOT use a handheld cell phone or other portable electronic device while operating the vehicle.
14. Drivers shall perform a pre-operational check of their vehicle.
15. Students are to stand clear when vehicle is been started.

Operation

1. Only one person shall operate the vehicle at a time. Ensure the area is clear of people and equipment before operating or working on it.
2. During operation and when cooling down, be aware that parts of the vehicle are hot.
3. When working in the engine bay and if the engine is running, stay clear of any moving parts including drivebelts or any moving shafts.
4. During operation, be aware when working on the cooling system, that it is under pressure and the coolant and water is hot. Ensure that you let it cool down before removing the radiator cap.
5. During operation and when working on the high-pressure lines of a petrol and diesel engine, depressurize the system first.
6. When charging a vehicle, ensure the car battery is ventilated and isolated from the vehicle.
7. Switch off the vehicle when work is completed.
8. Ensure the battery is turned off through the use of the green isolation switch when working on the vehicle.
9. Keep the equipment and work area safe, clean, and tidy.

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10. Follow the NZ Road Rules as defined by the NZ Transport Agency when driving the vehicle.
11. Ensure two-way communication with the tutor or another student before starting a vehicle
 - Person to shout out starting vehicle
 - The other person to shout out all clear
12. Ensure the engine bay is clear before starting the vehicle.
13. If necessary for busy times, use a person to direct-reversing vehicles – this person should be in visual contact with the driver at all times.
14. Toot the horn when entering or exiting the workshop to let anyone around you know there is a moving vehicle.
15. Drive defensively, observe speed limits, recognise potential hazards by scanning the road ahead and checking your rear-view and side mirrors. Expect the public to make mistakes.
16. Ensure appropriate PPE is used when working on vehicles.
17. When working on vehicle while engine is not operating, ensure key is removed from ignition and is put on the dash or seat.
18. When working on electric car, remove key and place on tutor's tool box, wear PPE.

Close Down

1. Switch off the vehicle when work is completed.
2. Put handbrake on.
3. Put window down when parked up inside.
4. Ensure key is out of the ignition and on the seat or dash.
5. Ensure battery isolation switch is switched off on training vehicles.

Routine maintenance

Lock out and/or isolate any equipment before inspection or maintenance.

	Name	Title	Signature	Date
Author	Elli Crawshaw	Area expert		07/02/2024
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Document Control

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